REV2026 · SEMESTER I · THEORY · 4 CREDITS

Mathematics for Computer Technology

Course Code 2001C · L:T:P:S 3:1:0:0 · 60 Contact Hours

Programme	AI, ML, Cloud, Cyber Forensics, CS, IT, Robotics etc.
Pre-requisite	1002 Fundamentals of Engineering Mathematics I
CIE	40
ESE	60
Self-Learning	60 hours

Official Source Declaration

Course: Mathematics for Computer Technology (2001C) · **Curriculum:** Kerala Polytechnic Revision 2026.

This handbook is prepared from the official syllabus PDF. All topics, subtopics, learning outcomes, and assessment rules are faithfully reproduced.

Verified Inconsistencies: The PDF lists 4 modules but Module I, II, III have 11 instructional hours each; Module IV has 12 hours (total 45 L + 15 T = 60). The self-learning table suggests 60 hours across 16 weeks. No contradictions detected.

Course Dashboard

Programme(s): Artificial Intelligence, ML, Cloud Computing, Cyber Forensics, CS, IT, Automation, Robotics, etc.

Subject Code: 2001C | **Semester:** I | **Type:** Theory

Teaching Scheme: 3:1:0:0 (L:T:P:S) | **Contact Hours:** 60

Credits: 4 | **CIE:** 40 | **ESE:** 60

Pre-requisite

Topic	Course Code	Course Title	Semester
Trigonometry, Differentiation, Matrices	1002	Fundamentals of Engineering Mathematics	I

Teaching and Assessment Scheme

Teaching/Week	Self-learning/Week	Total Hours/Sem	Credits	CIE	ESE	Total
3L+1T	5.5	60	4	40	60	100

How to Use This Handbook

1. **Understand** Read the module topics and deep explanations.
2. **Visualise** Study diagrams and interactive calculators.
3. **Apply** Solve worked examples and practice questions.
4. **Revise** Use quick revision cards and glossary before exams.

Course Outcomes & CO-PO Mapping

Course Outcomes (CO)

CO	Course Outcome	Bloom's Level
CO1	Apply concepts of complex numbers to solve problems.	Apply
CO2	Use integration techniques to solve basic engineering problems.	Apply
CO3	Perform data analysis using the foundation of logic and statistical tools.	Apply
CO4	Apply basic graph theory concepts to represent and solve real-world problems.	Apply

CO-PO Articulation Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	-	-	-	-	-	-	-	-	-
CO2	3	-	-	-	-	-	-	-	-	-	-
CO3	3	-	-	-	-	-	-	-	-	-	-
CO4	3	-	-	-	-	-	-	-	-	-	-
Legends: 1-Low, 2-Medium, 3-High, - No Correlation											

Curriculum Coverage Map

Module-wise topics, hours, CO, and anchor

Module	Title	Topics	Hours	CO	Anchor
I	Complex Numbers	System, operations, modulus, amplitude, polar form	11	CO1	#module-1
II	Integral Calculus	Reverse of differentiation, standard results, by parts, substitution, definite integrals, area	11	CO2	#module-2

III	Logic & Basic Statistics	Propositions, logical ops, truth tables, tautology, statistics, grouped data, central tendency	11	CO3	#module-3
IV	Graph Theory	Basic concepts, paths, walks, cycles, connectivity, trees, Dijkstra, Kruskal	12	CO4	#module-4

Coverage note: Graph classification includes the *complete graph*, in which every distinct pair of vertices is joined by an edge, in addition to null, simple, path, cycle, connected, regular and weighted graphs.

Module Roadmap

Module I
Complex Numbers → algebraic operations, polar form, Argand plane.

Module II
Integral Calculus → integration as anti-derivative, standard integrals, substitution, by parts, definite integrals.

Module III
Logic & Statistics → propositions, truth tables, tautology, grouped frequency, mean/median/mode.

Module IV
Graph Theory → vertices, edges, paths, cycles, trees, Dijkstra shortest path, Kruskal MST.

MODULE I

Complex Numbers

11 Hours CO1 Apply

Complex numbers extend the real number system to solve equations like $x^2 + 1 = 0$. They are essential in AC circuit analysis, signal processing, and control systems.

The Complex Number System, Conjugate, Equality

A complex number is expressed as **z = a + ib**, where **a** is the real part (Re z) and **b** is the imaginary part (Im z), and **i = √(-1)**.

Conjugate: The conjugate of z = a+ib is **$\bar{z}$ = a - ib**.

Equality: Two complex numbers are equal iff their real parts are equal and imaginary parts are equal.

Example: If $2x + 3iy = 4 + 6i$, then $2x = 4 \rightarrow x = 2$; $3y = 6 \rightarrow y = 2$.

Malayalam support: □□□□□□□□□ □□□□□□□□ (complex numbers) □□□□□□□□ (real), □□□□□□□□□ (imaginary) □□□□□□□□ □□□□□□□□□□□□. $i = \sqrt{-1}$ □□□.

Fundamental Operations (+, -, ×, ÷)

- **Addition:** $(a+ib) + (c+id) = (a+c) + i(b+d)$
- **Subtraction:** $(a+ib) - (c+id) = (a-c) + i(b-d)$
- **Multiplication:** $(a+ib)(c+id) = (ac - bd) + i(ad + bc)$
- **Division:** $(a+ib)/(c+id) = ((a+ib)(c-id))/(c^2+d^2)$

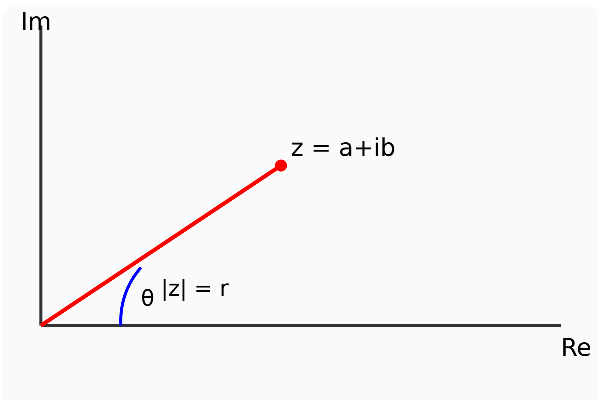
Example: $(3+2i)(1-4i) = 3\cdot1 - 2\cdot(-4) + i(3\cdot-4 + 2\cdot1) = 3+8 + i(-12+2) = 11 - 10i$.

Graphical Representation (Argand Plane), Modulus & Amplitude

Complex numbers are represented as points (a,b) in the Argand plane (x = real, y = imaginary).

Modulus (Absolute value): $|z| = \sqrt{a^2 + b^2}$.

Amplitude (Argument): $\theta = \tan^{-1}(b/a)$, measured in degrees or radians.



Argand plane: point (a,b) with modulus r and argument θ.

Complex Number Modulus & Argument

Real (a): 3 Imag (b): 4

Modulus: 5.00, Argument: 53.13°

Polar Form and Engineering Problems

The polar form of $z = a+ib$ is **$z = r(\cos\theta + i\sin\theta)$**, where $r = |z|$ and $\theta = \arg(z)$.

Example: Express $z = 1 + i$ in polar form. $r = \sqrt{(1+1)}=\sqrt{2}$, $\theta = \tan^{-1}(1/1)=45°$. So $z = \sqrt{2}\,(\cos 45° + i \sin 45°)$.

Engineering Application: Used in AC circuits for phasor representation, control systems, and signal processing.

MODULE II

Integral Calculus

11 Hours CO2 Apply

Integration is the reverse process of differentiation. It is used to find areas, volumes, and accumulated quantities in engineering.

Integration as Reverse of Differentiation, Standard Integrals

If $d/dx\,[F(x)] = f(x)$, then $\int f(x)\,dx = F(x) + C$.

- $\int x^n\,dx = x^{n+1}/(n+1) + C \quad (n\neq-1)$
- $\int e^x\,dx = e^x + C$
- $\int 1/x\,dx = \ln|x| + C$
- $\int \sin x\,dx = -\cos x + C$
- $\int \cos x\,dx = \sin x + C$

$\int \sec^2 x \, dx = \tan x + C$

Rules of Integration, Substitution (f(ax+b), f(x)·f'(x), f'(x)/f(x))

- **Constant multiple:** $\int k f(x) \, dx = k \int f(x) \, dx$
- **Sum/Difference:** $\int [f(x) \pm g(x)] \, dx = \int f(x) \, dx \pm \int g(x) \, dx$
- **Substitution:** For f(ax+b), substitute u = ax+b, du = a dx.
- For f(x)·f'(x), let u = f(x), du = f'(x) dx.
- For f'(x)/f(x), result is ln|f(x)| + C.

> **Example:** $\int (2x+3)^5 \, dx$. Let u = 2x+3, du = 2 dx → dx = du/2. = $1/2 \int u^5 \, du$ = $(1/2)(u^6/6) = (2x+3)^6/12 + C$.

Integration by Parts

Formula: $\int u \, dv = u v - \int v \, du$. Choose u as the function that simplifies on differentiation (LIATE rule).

> **Example:** $\int x \sin x \, dx$. Let u = x, dv = sin x dx → du = dx, v = -cos x. = $-x \cos x - \int (-\cos x) \, dx = -x \cos x + \sin x + C$.

Definite Integrals and Area Under Curve

Definite integral: $\int_a^b f(x) \, dx = F(b) - F(a)$.

Area bounded by curve and x-axis: $A = \int_a^b |f(x)| \, dx$.

> **Example:** $\int_0^1 x^2 \, dx = [x^3/3]_0^1 = 1/3$.

Definite Integral Calculator (x²)

a: 0 b: 1

Area = 0.3333

MODULE III

Logic and Basic Statistics

11 Hours   CO3   Apply

Propositions and truth tables form the basis of digital logic. Statistics helps in data analysis and decision making.

Propositions, Logical Operations (Negation, Conjunction, Disjunction, XOR)

Proposition: A statement that is either true or false.

- **Negation (¬p):** Opposite truth value.
- **Conjunction (p ∧ q):** True only if both are true.
- **Disjunction (p ∨ q):** True if at least one is true.
- **Exclusive OR (p ⊕ q):** True if exactly one is true.

Truth Table for Basic Operations

p	q	¬p	p∧q	p∨q	p⊕q
T	T	F	T	T	F
T	F	F	F	T	T
F	T	T	F	T	T
F	F	T	F	F	F

Tautology, Contradiction, Contingency

- **Tautology:** Always true (e.g., p ∨ ¬p).
- **Contradiction:** Always false (e.g., p ∧ ¬p).
- **Contingency:** Neither always true nor false.

Statistics: Grouped Data, Frequency Distribution, Measures of Central Tendency

Grouped data is organised into classes. **Frequency distribution** shows how often each value occurs.

Mean: $\bar{x} = (\Sigma f x) / \Sigma f$

Median: Middle value when data is ordered.

Mode: Most frequent value.

Example: Data: 2,3,5,5,7,8. Mean = (2+3+5+5+7+8)/6 = 5.0. Median = (5+5)/2 = 5. Mode = 5.

MODULE IV

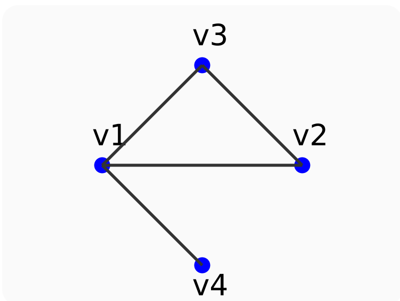
Graph Theory

12 Hours | CO4 | Apply

Graphs model networks, relationships, and paths. Used in computer networks, circuit design, and route planning.

Basic Concepts: Vertex, Edge, Degree, Loops, Simple Graph

- **Vertex:** Node or point.
- **Edge:** Line connecting two vertices.
- **Degree:** Number of edges incident to a vertex.
- **Loop:** Edge from a vertex to itself.
- **Simple graph:** No loops or multiple edges.



Simple graph with vertices v1-v4 and edges.

Handshaking Lemma, Subgraphs, Union of Graphs

Handshaking Lemma: Sum of degrees = 2 × (number of edges).

Subgraph: A graph formed from a subset of vertices and edges.

Union: Combined graph of two graphs with common vertices.

Matrix Representation: Adjacency and Incidence Matrices

Adjacency matrix: A $n \times n$ matrix where $A[i][j] = 1$ if edge exists, else 0.

Incidence matrix: $n \times m$ matrix where $B[i][j] = 1$ if vertex i is incident to edge j.

Types of Graphs: Null, Complete, Path, Cycle, Connected, Regular, Weighted, Tree, Spanning Tree

- **Null:** No edges.
- **Complete (K_n):** Every pair of vertices connected.
- **Path:** Linear sequence of vertices and edges.
- **Cycle:** Closed path.
- **Connected:** Path between every pair.
- **Regular:** All vertices have same degree.
- **Weighted:** Edges have weights.
- **Tree:** Connected acyclic graph.
- **Spanning tree:** Subgraph that is a tree and includes all vertices.

Dijkstra's Algorithm (Shortest Path) & Kruskal's Algorithm (Minimum Spanning Tree)

Dijkstra: Finds shortest path from source to all vertices in weighted graph.

Kruskal: Finds minimum spanning tree by sorting edges by weight and adding if no cycle.

> **Example (Kruskal):** Edges: AB(2), AC(3), BC(5), BD(1). Sorted: BD(1), AB(2), AC(3), BC(5). Add BD, AB, AC → MST weight = 1+2+3=6.

Formula Bank

Complex: $z = a+ib$, $|z| = \sqrt{a^2+b^2}$, $\arg z = \tan^{-1}(b/a)$, polar $= r(\cos\theta + i \sin\theta)$

Integration: $\int x^n\, dx = x^{n+1}/(n+1)$, $\int e^x\, dx = e^x$, $\int 1/x\, dx = \ln|x|$

By Parts: $\int u\, dv = uv - \int v\, du$

Definite: $\int_a^b f(x)\, dx = F(b) - F(a)$

Logic: $\neg(p \wedge q) = \neg p \vee \neg q$, $p \vee \neg p = T$

Statistics: Mean = $\Sigma fx/\Sigma f$, Median = middle value, Mode = most frequent

Graph: Sum of degrees = $2|E|$, Dijkstra, Kruskal MST

Visual Library

Argand Plane — complex number representation

Integration Area — area under curve

Graph Model — vertices, edges, trees

Self-Learning Activities

Module I: Find conjugate, modulus, amplitude, polar form of complex numbers. (16 hours)

Module II: Evaluate integrals using substitution and by parts; apply definite integrals. (13 hours)

Module III: Construct truth tables; solve mean/median/mode problems; applications of statistics. (14 hours)

Module IV: Draw real-life graphs; identify paths/cycles; find applications of graph theory. (17 hours)

Assessment Methodology

CIE and ESE Breakdown

Mode	Attendance	CA1	CA2	CA3	CA4	CA5	CA6	ESE
Marks	5	15	15	15	15	50	60	60
Converted	5	15			20			60
Total	CIE 40 + ESE 60 = 100							

Series Examination Pattern: 2 hours, any one module. Part A (2×3=6), Part B (3×3=9), Part C (5×4=20) — total 35 marks.

Module-wise Questions with Answers

Module I

1. Define complex number and find the conjugate of 3+4i.

A complex number z = a+ib. Conjugate = 3-4i.

2. Express z = 1 - i in polar form.

$r=\sqrt{2}$, $\theta=-45°$, $z=\sqrt{2}(\cos 45° - i \sin 45°)$.

Module II

3. Evaluate $\int (3x^2+2x)\,dx$.

$x^3 + x^2 + C$.

4. Find area under $y = x^2$ from x=0 to 2.

$\int_0^2 x^2\,dx = [x^3/3]_0^2 = 8/3$.

Module III

5. Construct truth table for p ∧ q.

T&T→T, T&F→F, F&T→F, F&F→F.

6. Find mean of data: 2,4,6,8,10.

Mean = 6.

Module IV

7. What is a tree? Give example.

A connected acyclic graph. Example: family tree.

8. State Handshaking Lemma.

Sum of degrees = 2 × number of edges.

Practice Model Paper

Based on the official pattern, not an official SITTTR model question paper.

Part A (2 marks each)
1. Define complex number. 2. What is the conjugate of 5-2i? 3. Write the integral of 1/x. 4. Define tautology.

Part B (3 marks each)
5. Express 2+2i in polar form. 6. Evaluate ∫ x sin x dx. 7. Construct truth table for p∨q.

Part C (5 marks each, with choice)
8. a) Find modulus and argument of -1+i. OR b) Evaluate $\int_0^1 (x^2+x)\,dx$. 9. a) Apply Kruskal's algorithm to find MST. OR b) Explain Dijkstra's algorithm.

Quick Revision

Module I $i^2=-1$, $|z|=\sqrt{a^2+b^2}$, polar r(cosθ+i sinθ)

Module II $\int x^n \, dx = x^{n+1}/(n+1)$, by parts: $\int u \, dv = uv - \int v \, du$

Module III Truth tables, mean=$\Sigma fx/\Sigma f$, median=middle, mode=most frequent

Module IV Graph = (V,E), tree = connected acyclic, MST via Kruskal

Common Mistakes: Forgetting +C in indefinite integrals; mixing real/imaginary parts; incorrect truth table for XOR.

Exam Checklist: Revise formulas, practice truth tables, solve at least one MST and shortest path problem.

Glossary

Term	Meaning
Complex Number	Number of form a+ib
Modulus	Absolute value, distance from origin
Amplitude	Angle of complex number with real axis
Integral	Anti-derivative, area under curve
Proposition	Statement with truth value
Tautology	Always true
Contradiction	Always false
Graph	Set of vertices and edges
Tree	Connected acyclic graph
Spanning Tree	Tree containing all vertices

Learning Resources

Textbook: Mathematics for Computer Technology, SITTTR Kalamassery.

Web Resources: (None listed in PDF)